Adaptive Hospital Readmission Decision Support Framework (2026)

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Abstract

Readmission is a critical area of concern for the hospital and current interventions include standard discharge practices and clinical judgement of patient records to provide probabilistic readmission. Alternative tested methods have proven to be effective on a small scale but failed on larger trials. The proposed framework is a clinical decision support dashboard that incorporates visualization of patient data on a high and granular level, a readmission prediction tool using machine learning accompanied by the display of contributing risk factors and care recommendations while supporting clinicians discharge decisions and planning. The framework segregates the information across multiple dashboard views to provide insights unique to each role while maintaining consistency in the information provided.

Introduction

Through the lens of a Patient's Hospital Journey

A patient visits the ER and is admitted to the hospital for treatment. Post treatment, when the patient is stable enough, the clinicians will decide the patient is stabilized for discharge. Hospitals generally follow a similar procedure; the main emphasis is on the medications and/or care the patient will be required to continue and the length of time after. The more complex the diagnosis, the more complex the discharge guidelines are expected to be. Some patients may already be anticipating readmission due to planned future hospital care such as treatment, surgical procedures or further assessment for more thorough diagnosis. Due to the severity, patients may be required to recoup from the treatments provided during the initial admission or may require out-patient monitoring prior to readmission for the proposed health care plan. In other cases, a readmission risk assessment tool may be utilized to predict the probability of readmission. Clinical judgement may also be employed to determine the potential risk. This risk score may be accompanied by a predicted time frame until the next readmission. However, this assessment is not always accurate. The reasons for readmission may include low efficacy of prescribed treatment, insufficient or misaligned diagnosis or post-care guidelines or due to the patient's lack of adherence to post-discharge instructions. Readmission may also be the result of unanticipated and unavoidable worsening of patient conditions.

Literature Review

The Readmission Challenge

Hospitals face significant challenges in reducing readmissions such as limited resources, difficulty in addressing the uniqueness of patient needs and insufficient control of post hospital care [5]. Patients often require follow-up treatments post discharge and may fail to be consistent due to the cost of medications, progression of illnesses, development of new illnesses, inadequate ability to care for oneself, or to receive assistance, health illiteracy or their lack of awareness of their diagnosis or potential progression and severity. Lower income individuals, persons with disabilities and persons who are not health conscientious may be unable to adequately care for themselves after discharge and are likely to face relapse, worsening conditions or the development of secondary conditions. As such, reducing readmission becomes more difficult to manage by hospitals. Furthermore, patients may visit other hospital facilities for readmission thus reducing the capability of tracking the true hospital readmission rate and assessing treatment outcomes. The typical discharge process includes disseminating prescribed medication instructions, wound care instructions, specialist follow-up appointment schedule, medical equipment requirements and follow-up or continued care guidelines as applicable [2]. This provides the necessary information for patients to manage their health while resuming their normal or adapting to a new normal lifestyle.

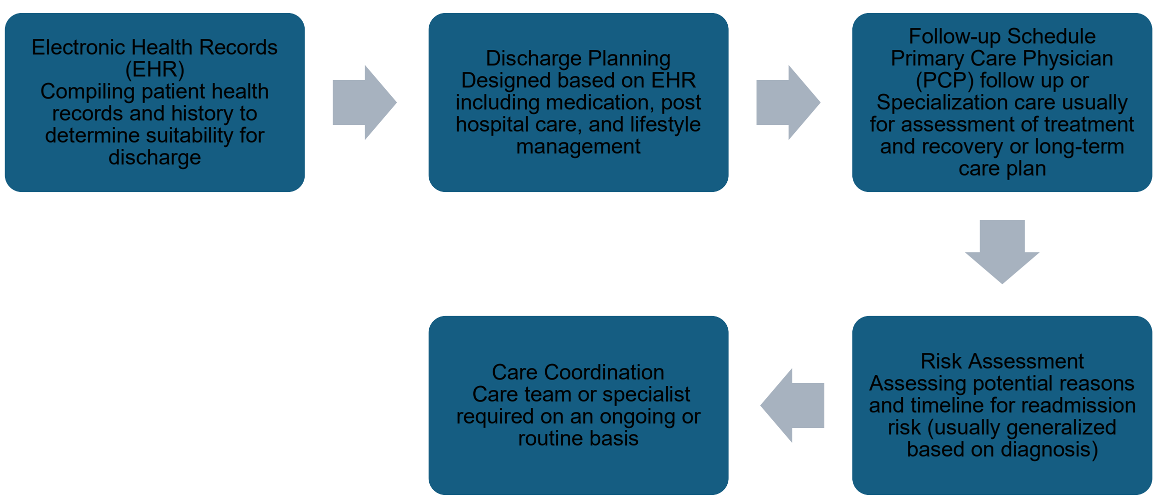
Readmission as an Opportunity for Intensified Treatment

The average hospital readmission rate is 14.67% [2]. While this may appear low, it bears significant hospital costs, patient expenses, welfare and health risk. Analysis presents that higher readmission is typically associated with older age, chronic conditions and intensive care [2]. These indicate factors that will increase the difficulty of reducing readmission rate. The senior population are usually less capable of caring for oneself, may forget medication schedules, have limited resources to maintain follow-up appointments and care and are hence at greater risk for readmission. Chronic conditions present with ongoing need for management and flare-ups can result in the need for continuous treatment. Individuals with chronic conditions, especially those with multiple comorbidities, are at greater risk for disease progression and as such may need ongoing periodic care. Furthermore, patients who undergo intensive care may require readmission dependent on the recovery process which varies on the patient's previous health history and family health history. These are some scenarios that underscores the value in readmission as it presents the opportunity for clinicians to reevaluate patients, perform further assessments based on patient feedback and provide more insight on patient diagnosis. Furthermore, there is the opportunity to explore alternative treatments based on the outcomes of earlier phases of treatment as well as to monitor patients while administering treatment and post treatment for adverse reactions. Readmission may also be an opportunity for recurring patients to alleviate symptoms, receive medical care that is not possible in outpatient and resume their daily activities post discharge. Therefore, this indicates that readmission may be a requirement for some patients, and monitoring this may offer greater insights and ability to prepare hence reducing cost and providing improved treatment.

The Hospital Readmissions Reduction Program

Nevertheless, the introduction of the Hospital Readmissions Reduction Program, a Medicare purchasing program, was designed to reduce avoidable readmissions by improving communication and coordination of post-discharge planning [1]. This program considers the readmission during the first 30 days after the initial discharge for the same or alternative diagnosis excluding some planned readmission [1]. This 30-day timeframe is considered, as readmissions during such a short time frame may be attributed to poor treatment strategies during the initial admission which indicates avoidable readmission prompting hospitals to conduct more thorough assessments and ensuring patient stability prior to discharge. This program emphasizes the need to minimize preventable readmissions through greater emphasis on treatment outcomes rather than optimizing throughput.

The Standard Readmission Process



The current process is much more generalized and does not emphasize the uniqueness of each patient's clinical history, economic standards, support system and admission diagnosis to fully understand and generate a risk prediction or post care management system. As such, this may not fully capitalize on the opportunity to reduce readmission and increase patient outcomes. Integrating direct patient vitals and results to generate a profile may offer greater insight into potential readmission.

Current Solutions to Reducing Readmission

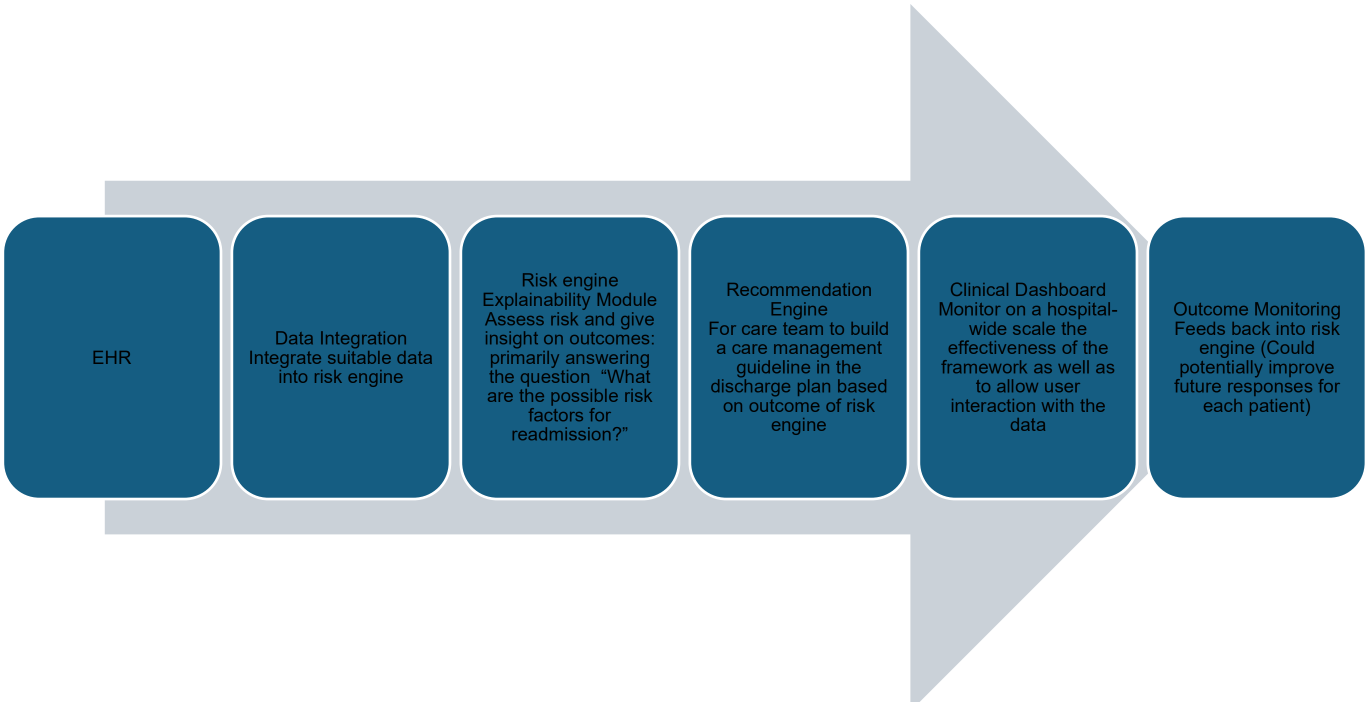
Current approaches to reducing hospital readmissions include care transition programs that support patients as they transition from inpatient to outpatient care, medication reconciliation to ensure patients understand the purpose of each medication and the prescribed dosages, post-discharge follow up, patient education, care coordination and addressing the social determinants of health to reduce the barriers to receiving care and managing daily activities [3]. These steps represent standard clinical practices aimed at supporting patients' continued care post discharge and reducing preventable readmissions. However, these interventions do not identify high risk patients and the reason for the elevated risks. According to [6], telehealth monitoring exemplifies another approach to reducing readmission rates, however, a large trial showed no reduction. This suggests that monitoring alone may be insufficient to address the complexity of readmission risk. Patients at higher risk often exhibit characteristics such as advanced age, polypharmacy and decreased functional status while using predictive models that quantify the risk may create an opportunity to identify potential readmission [6]. By using a probabilistic approach to patient care, clinicians can ascertain the likelihood of readmission, and the clinical variables that contribute to the risk to coordinate more proactive care.

Problem Statement

These limitations in addressing readmission risk motivated the development of predictive support systems that can be utilized by various stakeholders within the healthcare setting to estimate readmission risk, provide clinically interpretable insights and personalized care recommendations as well as to monitor patient risk over time. This system facilitates a more patient centered discharge planning through identifying the appropriate resources and follow-up care required.

Proposed Solution

The Proposed Adaptive Readmission Reduction framework:

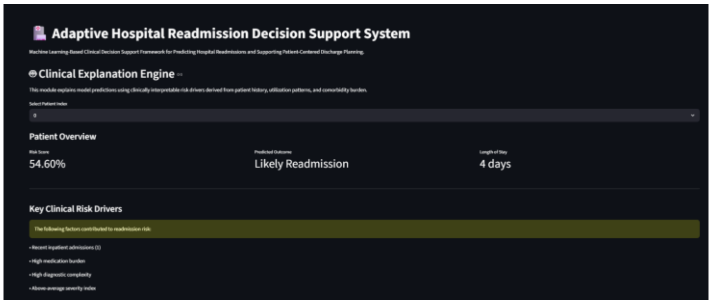


The proposed framework utilizes an Adaptive Readmission Decision Support Dashboard that provides visualizations for clinicians and administrators while also functioning as a patient-level risk assessment and care management tool. The framework combines predictive analytics with clinical decision support by estimating readmission risk, explaining the factors contributing to the risk and generating care recommendations while monitoring outcomes over time. The dashboard uses a trained Random Forest machine learning model to estimate the probability of a 30-day hospital readmission. This predicted risk, combined with rule-based clinical logic, is used to generate personalized discharge recommendations.

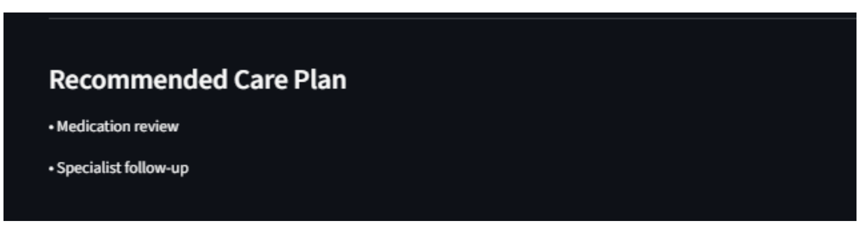
The framework begins with data integration where relevant patient information is extracted from the Electronic Health Record to build the patient's profile. This information includes:

- disease diagnosis
- demographic
- medical history
- laboratory results
- previous admissions
- social factors
- discharge summaries

This information feeds into the risk engine to estimate the probability of readmission and provide potential factors for readmission. This model predicts, using Random Forest machine learning model, the risk over a 30-day period aligning with standard clinical readmission measures. For a more clinically usable model, the most influential predictors contributing to the readmission risk at the population level were identified using feature importance.



The output from the risk engine then feeds into the recommendation engine that provides patient care suggestions to clinicians during discharge planning. These could include medication review, specialist follow up or post-discharge monitoring. This aids in identifying the appropriate care team members required during discharge. Depending on the identified risk factors, recommendations may indicate the need for a nurse to provide care education or a social worker to coordinate community resources, transportation services, obtain supportive medical equipment or home-based support. The recommendation engine is powered by rule-based clinical logic that combines the predicted risk generated by the random forest model with selected patient characteristics such as medication burden, history of hospitalization and pre-existing illnesses to generate a recommended personalized post-discharge plan. This serves as support rather than a replacement of clinical judgement.



The risk engine and recommendation engine are support tools that will function as clinical support systems and provide clinicians greater flexibility and opportunities for more patient-centered actionable care management suggestions. Given the influx of patients that may reduce the length of time a clinician has with a patient, this framework maximizes that time, where clinicians can maximize their time on health care diagnosis while the tool aids in the prediction system and care management.

To support decision-making across various units, the framework includes several components within the clinical dashboard. The information is presented strategically to provide multiple perspectives to ensure suitable interpretation according to each healthcare staff role while maintaining consistency across the information provided.

Executive Dashboard

The executive dashboard provides a broad overview of patient risk statistics, contributing factors and proposed management strategies. This view is primarily intended for hospital administrators and leadership to view operational performance and patient risk at a high- level.

Hospital Analytics

The Hospital Analytics dashboard provides more detailed insights for social workers, nurses, pharmacists and other care coordinators. This dashboard includes:

Population risk: Distribution of patients across the risk categories: low, moderate and high.

Diagnoses: Analysis of the risk classification of major comorbidities including the comparative demonstration of readmission risk for patients with and without diabetes.

Medications: Assessment of the relationship between medication and predicted readmission risk as well as the medication burden and distribution across the patient population.

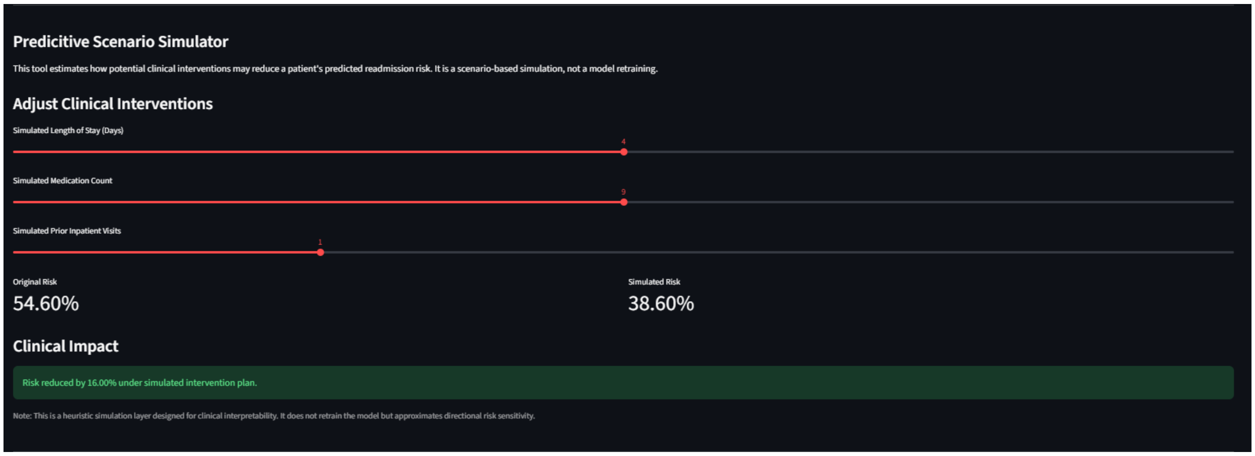
Recommendations: Summary of the most frequently generated recommendation strategies across the patient population.

Hospital Operations

The hospital operations dashboard, through the identification of patient distribution across departments, outcome and discharge recommendations, aims to improve resource utilization, staffing and operational performance.

Predictive Simulator

In addition to the predictive tool, the framework incorporates a Predictive Scenario Simulator that allows clinicians to modify selected patient dimensions to observe the estimated impact on predicted readmission risk. This interactive feature supports clinicians in assessing how targeted interventions can influence outcomes prior to discharge.



Potential Users of the framework

The proposed framework supports several healthcare stakeholders including:

- Clinicians: Understand readmission risk and drivers to support discharge planning and care strategies
- Pharmacist: Review potential medication adherence concerns
- Nurse: Tailor discharge education including explanation of medication schedule, care management and follow-up appointments.
- Case Managers and Social Workers: Coordinate support systems, resources and home support services.
- Hospital Administrators: Monitor outcomes, readmission trends, staffing requirements and resource utilization.

The current framework is designed as a prototype using manually updated patient data. However, the architecture supports the integration with Electronic Health Record systems or other databases to enable automated data transfer and model updates. A risk threshold of 0.35 is used as this threshold demonstrates an appropriate trade-off and balance between false negatives and false positives. The dashboard allows for variation of this threshold to compare outcomes.

The aim of the revised framework is to give greater insights into the potential reasons for readmission to potentially shift from a reactive process to providing risk management and preventative care. By combining predictive analytics, interpretable decision support, personalized care recommendations and operational monitoring within the framework, clinicians and administrators are provided with greater insight into readmission factors and interventions to reduce that risk.

The framework is to be considered a conceptual framework rather than a validated clinical support framework. Its implementation requires strategic cooperation amongst various clinicians, health care workers and administrators to collect required data and disseminate the appropriate information. Ultimately, the framework aims to reduce hospital readmission rates and associated costs while improving patient outcomes through more support for a patient-centered care.

Conclusion

The significant challenges of reducing readmission rates along with the potential consequences of readmission for both the patient and hospital indicates the need for a strategic intervention. Clinical judgement of presented risk factors such as age, disease progression, admission history and comorbidities serves as the primary tool for calculating the probability of readmission. The proposed framework visualizes the data, predicts readmission, explains the factors contributing to the probability and supports clinical discharge planning through suggestion of care strategies. Furthermore, it expands its usability for other care team members and administrators through the presentation of high-level summaries and detailed granular data. Additionally, the framework provides a simulation tool to support the discharge decision in terms of deciding if alternative treatment methods may reduce the predicted readmission. The framework serves as a support tool rather than a replacement of clinical judgment and requires the collaborative effort of all care team members for its effectiveness. Although this prototype requires manual intervention for data ingestion, its architecture supports future expansion for more automation and continuous updates.

Appendix

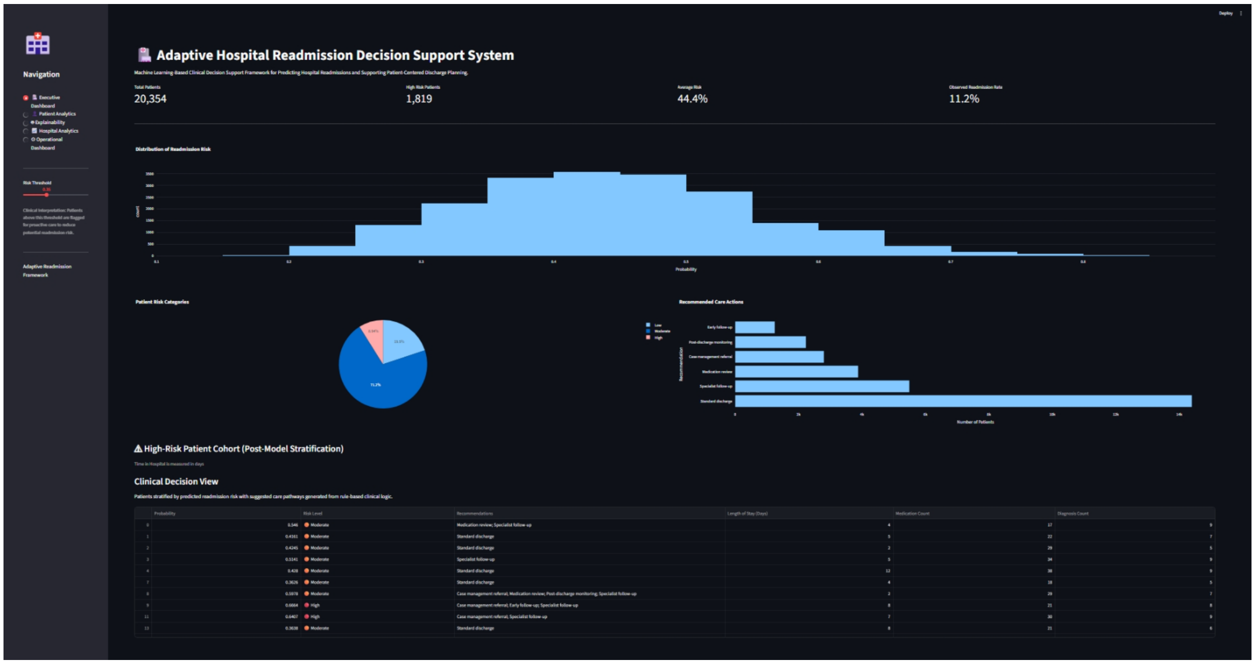


Figure 1. Executive Dashboard displaying high-level summary data

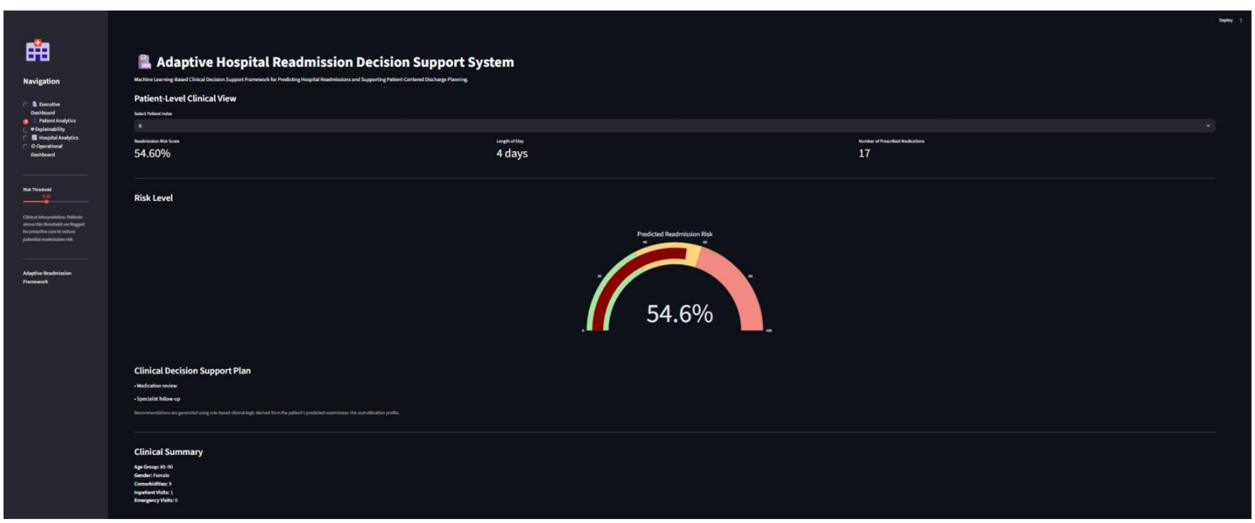


Figure 2 a. Patient Analytics Risk Level, Clinical Summary and Decision Support Plan

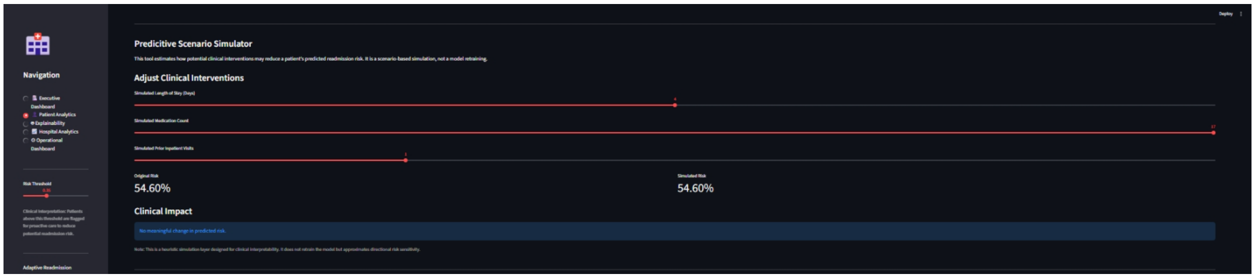


Figure 2 b. Predictive Scenario Simulator to Compare How Various Intervention Strategies Affect the Readmission Risk Score

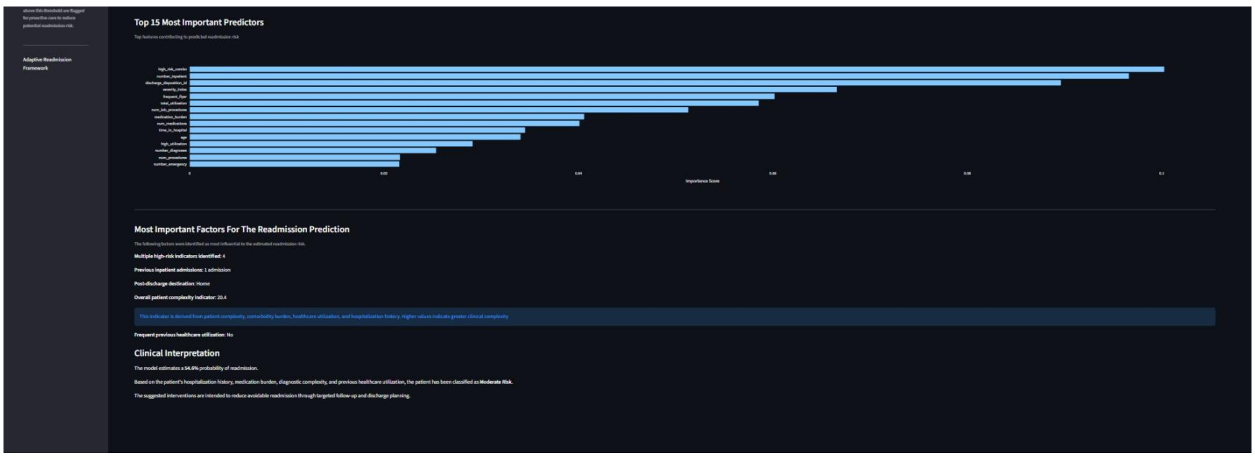


Figure 2 c. Insights into the Features that Impact the Readmission Probability at the Population Level

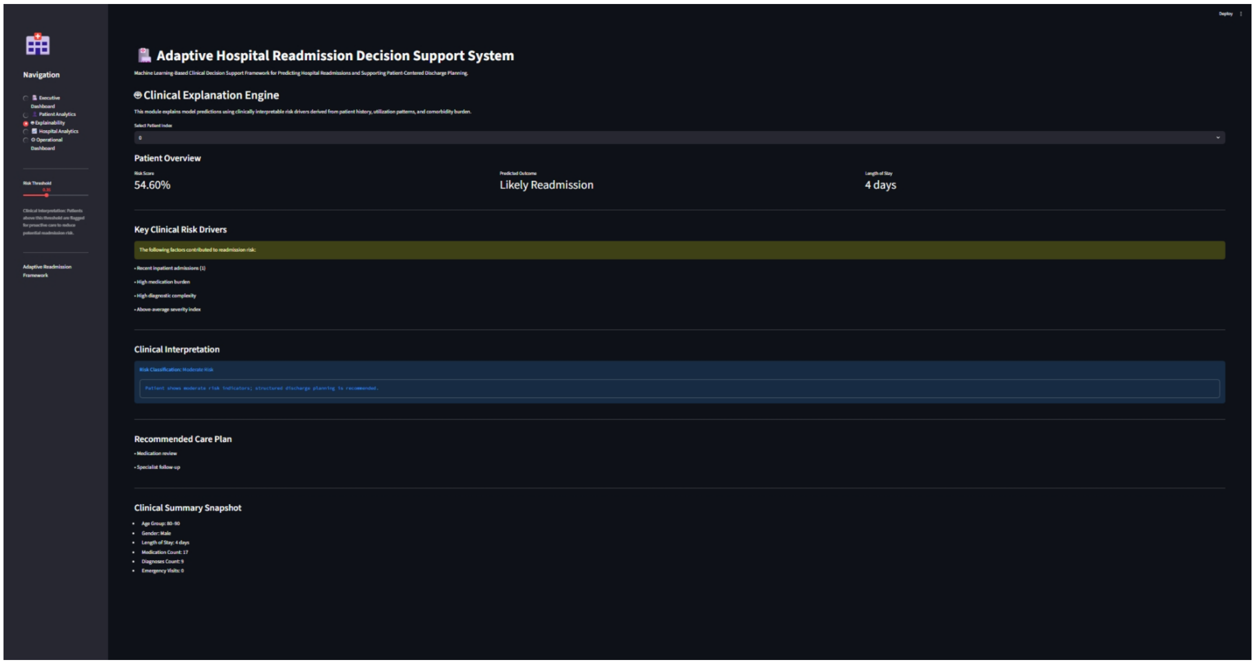


Figure 3. Patient Explainability Dashboard – An Overview of the Readmission Prediction, Risk Drivers and Care Recommendations

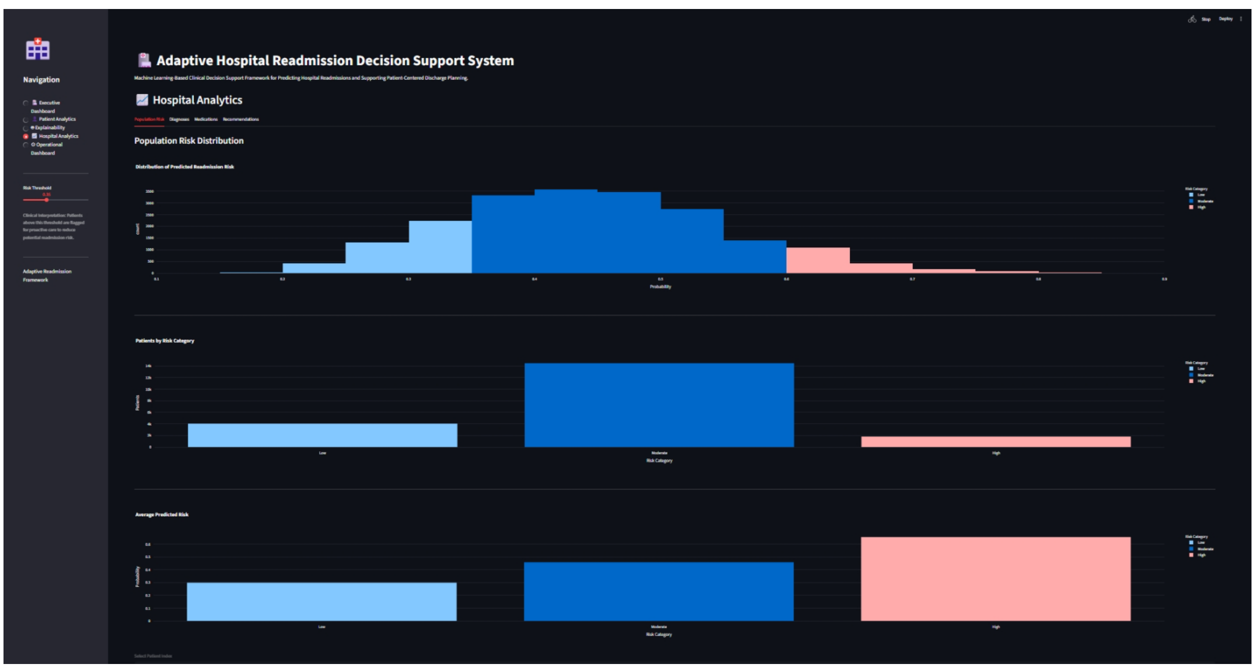


Figure 4 a. Hospital Analytics – Population Risk Overview

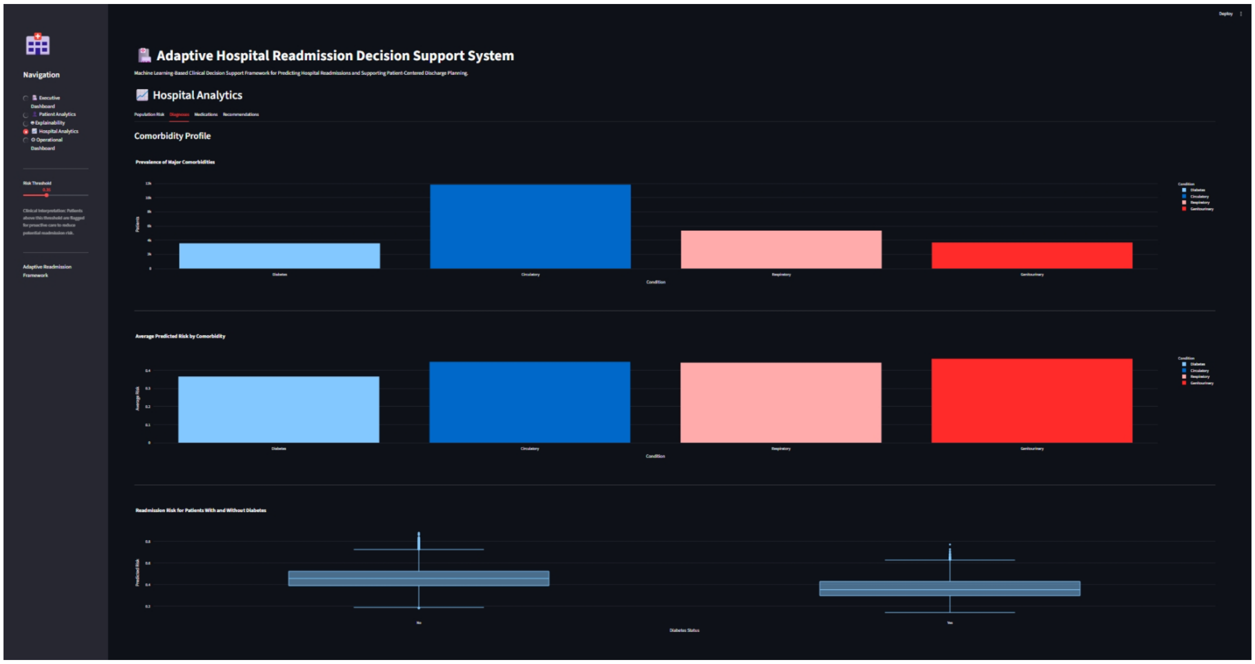


Figure 4 b. Hospital Analytics – Diagnosis Overview

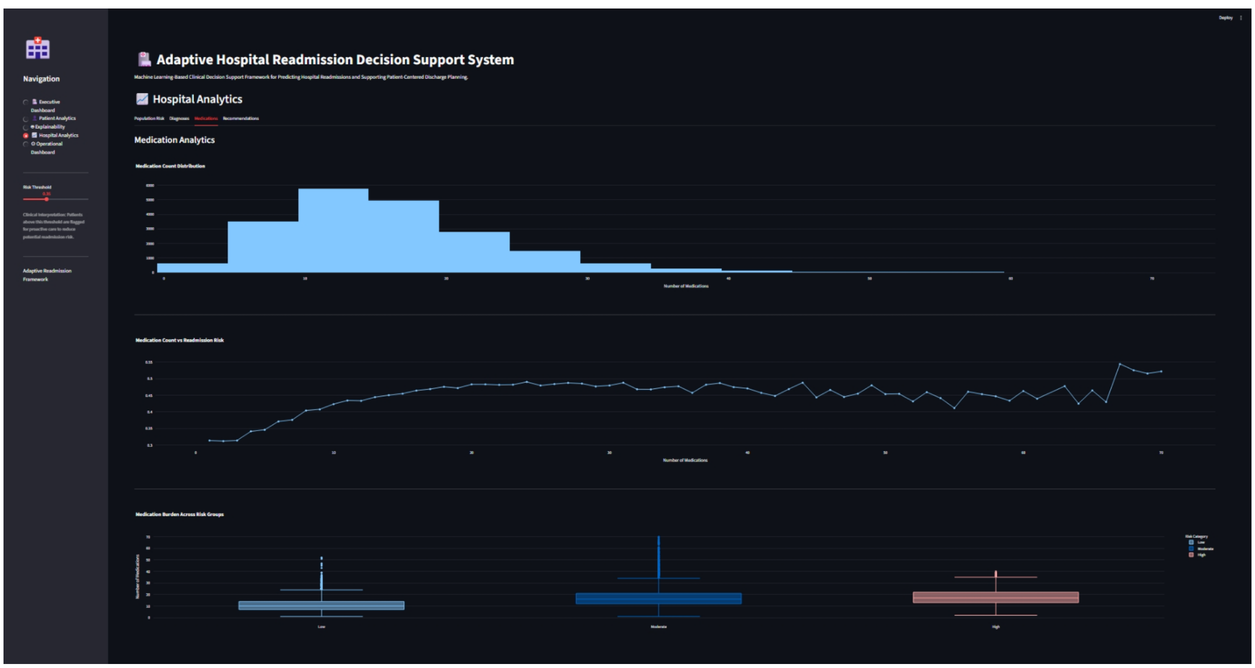


Figure 4 c. Hospital Analytics – Medication Analysis

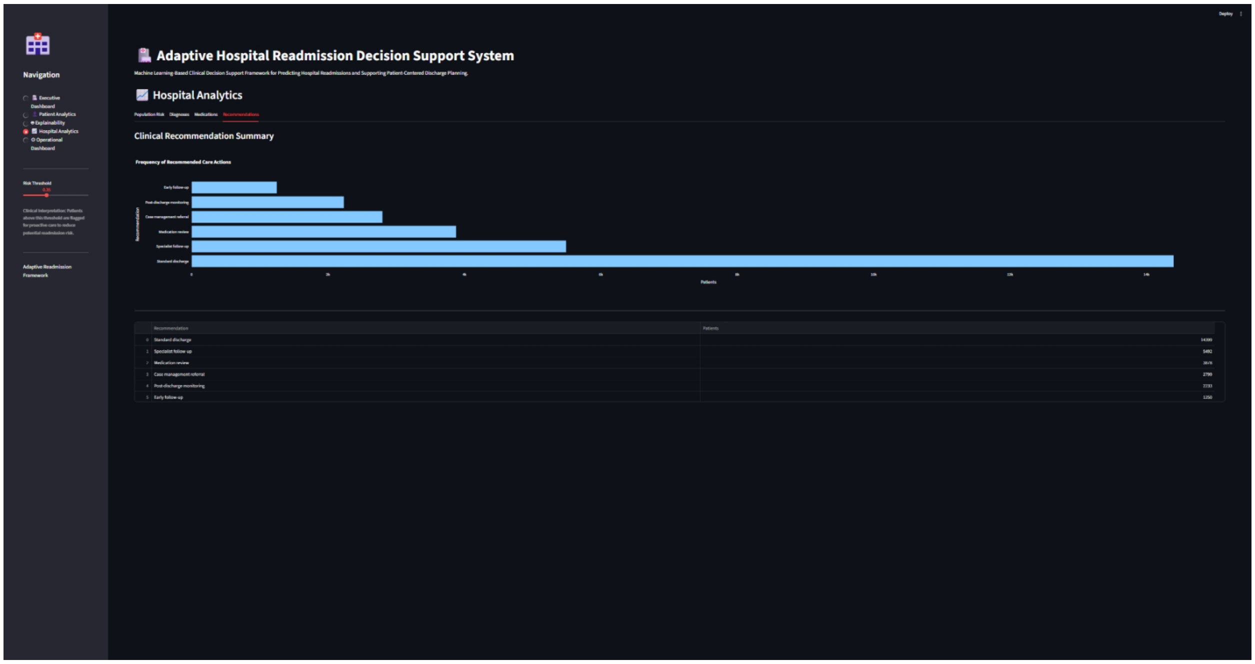


Figure 4 d. Hospital Analytics – Recommendation Summary

Figure 5 a. Administrative and Operational Dashboard – Operational Assessment

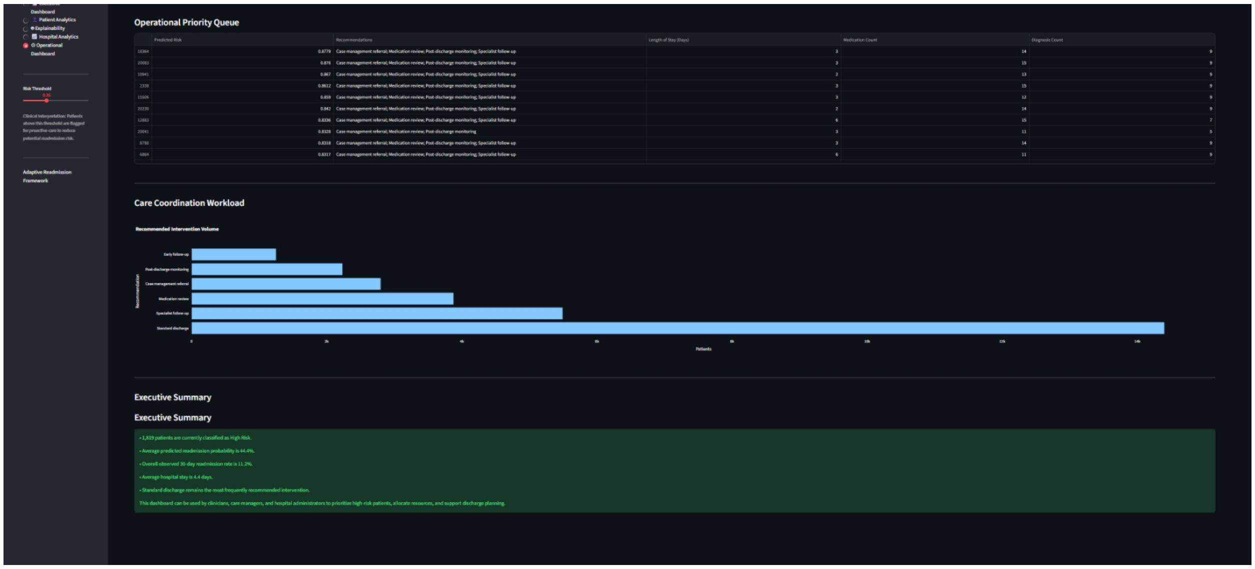


Figure 5 b. Administrative and Operational Dashboard – Clinical Assessment

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